Root Cause Analysis Report

Part	MS-4505-001 Sheet Metal Bracket
Material / process	SPCC Steel / bending
Case	rca_0493bc1ddc164a189398e1025b4af702
Date	2026-11-01 17:54:08
Status	AI proposal, pending expert review

WARNING: the AI model was not used for this analysis (fallback-fake (gemini failed: JSONDecodeError: Expecting ',' delimiter: line 49 column 49 (char 2511))). The text below is a generic offline template, not a result.

1. Problem

Defect	Severity	Confidence	Detail
size_mismatch	major	99%	length 118.44 mm (nom 120.00 ±0.10)
size_mismatch	major	99%	hole dia 12.73 (nom 12.00 ±0.05)
size_mismatch	major	99%	hole position off 1.91 mm (±0.10)

2. Root cause

Cause: Improper Bending Force / Over Bending

Reason: The detected deformation and dimensional inaccuracy strongly suggest excessive or improperly calibrated bending force. Retrieved RCA documents confirm this pattern for SPCC steel bending operations.

Effect: Deformation, Dimensional Inaccuracy

Confidence: 70%

Supporting evidence
- **seed-001**: Improper Bending Force or Over Bending causes deformation and dimensional inaccuracy. Precautions: optimize bending parameters, use proper tooling, calibrate machines regularly.
- **seed-004**: Springback in sheet metal bending causes dimensional inaccuracy. Overbend compensation and material property testing are required.

3. Solution

Corrective actions (fix this occurrence)

1. Re-set the press-brake bending force and angle to the material specification
2. Re-inspect the affected lot for deformation and dimensional deviation

Preventive actions (stop it recurring)

1. Add bending-force verification to the machine calibration schedule
2. Control bending parameters per material batch

Precautions

- Optimize bending parameters to material specification
- Use proper tooling and punch/die sets
- Regular machine calibration and force verification

How to verify the fix

Re-inspect the next 30 parts; no deformation and all dimensions within tolerance

4. Other causes considered

Rank	Cause	Confidence	Reason
2	Material Inconsistency / Hardness Variation	67%	Hardness variation in the incoming material can cause unpredictable springback and cracking. Material certificate verification is required.
3	Tool Wear / Die Damage	64%	Progressive die wear causes scratches, dents, and edge burrs. Preventive maintenance schedule should be implemented.

5. Traceability

AI model: fallback-fake (gemini failed: JSONDecodeError: Expecting ',' delimiter: line 49 column 49 (char 2511)). Expert edited the cause: no.

This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.